

Mocking Dependencies in Spring Tests using Mockito

1. pom.xml

Add these dependencies:

```
<dependencies>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-data-jpa</artifactId>
    </dependency>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
    </dependency>

    <dependency>
        <groupId>com.h2database</groupId>
        <artifactId>h2</artifactId>
        <scope>runtime</scope>
    </dependency>

</dependencies>
```

2. Spring Boot Main Class

DemoApplication.java

```
package com.example.demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class DemoApplication {
```

```
public static void main(String[] args) {  
    SpringApplication.run(  
        DemoApplication.class,  
        args);  
}  
}
```

3. User Entity

User.java

```
package com.example.demo.entity;
```

```
import jakarta.persistence.Entity;  
import jakarta.persistence.Id;
```

```
@Entity
```

```
public class User {
```

```
    @Id
```

```
    private Long id;
```

```
    private String name;
```

```
    public User() {  
    }  
}
```

```
    public User(Long id, String name) {  
        this.id = id;  
        this.name = name;  
    }  
}
```

```
    public Long getId() {  
        return id;  
    }  
}
```

```
    public void setId(Long id) {  
        this.id = id;  
    }  
}
```

```
    public String getName() {
```

```
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
}
```

4. Repository

UserRepository.java

```
package com.example.demo.repository;

import org.springframework.data.jpa.repository.JpaRepository;

import com.example.demo.entity.User;

public interface UserRepository
    extends JpaRepository<User, Long> {

}
```

5. Service

UserService.java

```
package com.example.demo.service;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import com.example.demo.entity.User;
import com.example.demo.repository.UserRepository;

@Service
public class UserService {

    @Autowired
    private UserRepository userRepository;

    public User getUserById(Long id) {

        return userRepository
            .findById(id)
    }
}
```

```
        .orElse(null);
    }
}
```

6. Controller

UserController.java

```
package com.example.demo.controller;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;

import com.example.demo.entity.User;
import com.example.demo.service.UserService;

@RestController
@RequestMapping("/users")
public class UserController {

    @Autowired
    private UserService userService;

    @GetMapping("/{id}")
    public ResponseEntity<User>
        getUser(@PathVariable Long id) {

        return ResponseEntity
            .ok(userService.getUserById(id));
    }
}
```

Exercise 1

UserControllerTest.java

Mocking Service Dependency in Controller Test

```
package com.example.demo;

import static org.mockito.Mockito.when;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.*;
```

```
import com.example.demo.controller.UserController;
import com.example.demo.entity.User;
import com.example.demo.service.UserService;
```

```
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.autoconfigure.web.servlet.WebMvcTest;
import org.springframework.boot.test.mock.mockito.MockBean;
import org.springframework.test.web.servlet.MockMvc;
```

```
@WebMvcTest(UserController.class)
public class UserControllerTest {
```

```
    @Autowired
    private MockMvc mockMvc;
```

```
    @MockBean
    private UserService userService;
```

```
    @Test
    void testGetUser()
        throws Exception {
```

```
        User user =
            new User(1L, "John");
```

```
        when(userService
            .getUserById(1L))
            .thenReturn(user);
```

```
        mockMvc.perform(
            get("/users/1"))
            .andExpect(
                status().isOk())
            .andExpect(
                jsonPath("$.id")
                    .value(1))
            .andExpect(
                jsonPath("$.name")
                    .value("John"));
```

```
}  
}
```

Exercise 2

UserServiceTest.java

Mocking Repository in Service Test

```
package com.example.demo;  
  
import static org.junit.jupiter.api.Assertions.*;  
import static org.mockito.Mockito.*;  
  
import java.util.Optional;  
  
import com.example.demo.entity.User;  
import com.example.demo.repository.UserRepository;  
import com.example.demo.service.UserService;  
  
import org.junit.jupiter.api.Test;  
import org.junit.jupiter.api.extension.ExtendWith;  
  
import org.mockito.InjectMocks;  
import org.mockito.Mock;  
import org.mockito.junit.jupiter.MockitoExtension;  
  
@ExtendWith(MockitoExtension.class)  
public class UserServiceTest {  
  
    @Mock  
    private UserRepository repository;  
  
    @InjectMocks  
    private UserService service;  
  
    @Test  
    void testGetUserById() {  
  
        User user =  
            new User(1L,  
                "John");
```

```
        when(repository
            .findById(1L))
            .thenReturn(
                Optional.of(user));

        User result =
            service.getUserById(1L);

        assertNotNull(result);

        assertEquals(
            "John",
            result.getName());

        verify(repository)
            .findById(1L);
    }
}
```

Exercise 3

UserIntegrationTest.java

Integration Test with Mocked Service

```
package com.example.demo;

import static org.mockito.Mockito.when;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.*;

import com.example.demo.entity.User;
import com.example.demo.service.UserService;

import org.junit.jupiter.api.Test;

import org.springframework.beans.factory.annotation.Autowired;

import org.springframework.boot.test.autoconfigure.web.servlet.AutoConfigureMockMvc;

import org.springframework.boot.test.context.SpringBootTest;

import org.springframework.boot.test.mock.mockito.MockBean;
```

```
import org.springframework.test.web.servlet.MockMvc;
```

```
@SpringBootTest
```

```
@AutoConfigureMockMvc
```

```
public class UserIntegrationTest {
```

```
    @Autowired
```

```
    private MockMvc mockMvc;
```

```
    @MockBean
```

```
    private UserService userService;
```

```
    @Test
```

```
    void testIntegration()
```

```
        throws Exception {
```

```
        User user =
```

```
            new User(
```

```
                1L,
```

```
                "John");
```

```
        when(userService
```

```
            .getUserById(1L))
```

```
            .thenReturn(user);
```

```
        mockMvc.perform(
```

```
            get("/users/1"))
```

```
            .andExpect(
```

```
                status().isOk())
```

```
            .andExpect(
```

```
                jsonPath("$.id")
```

```
                    .value(1))
```

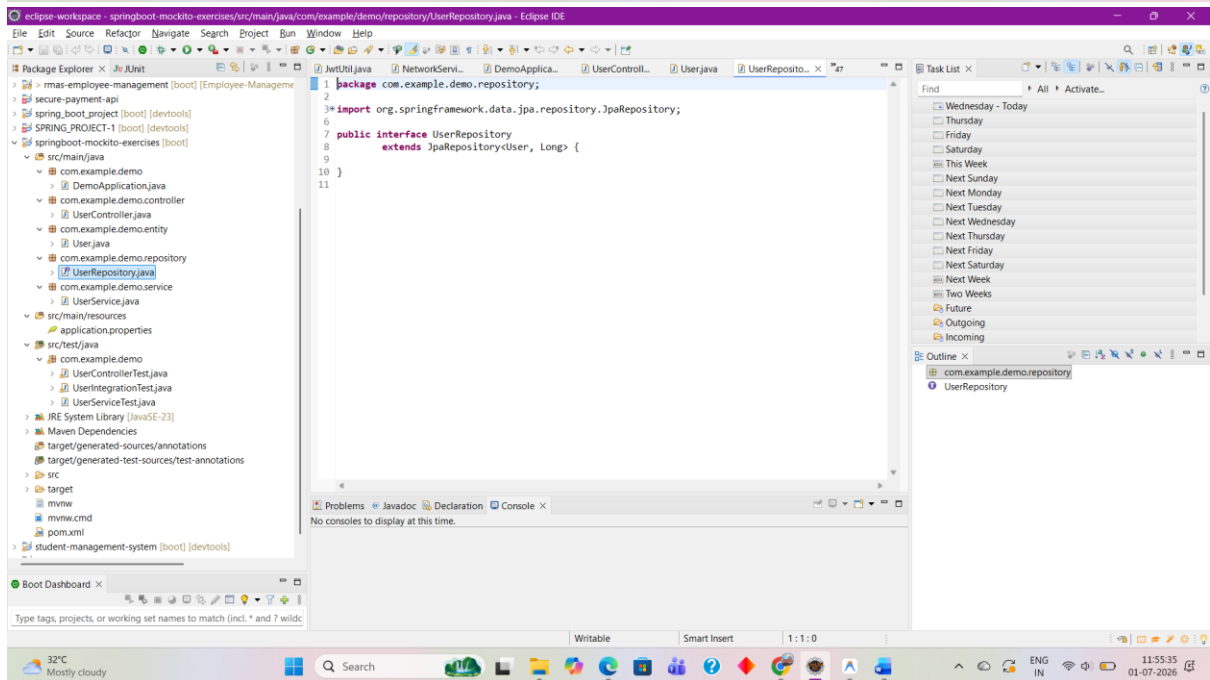
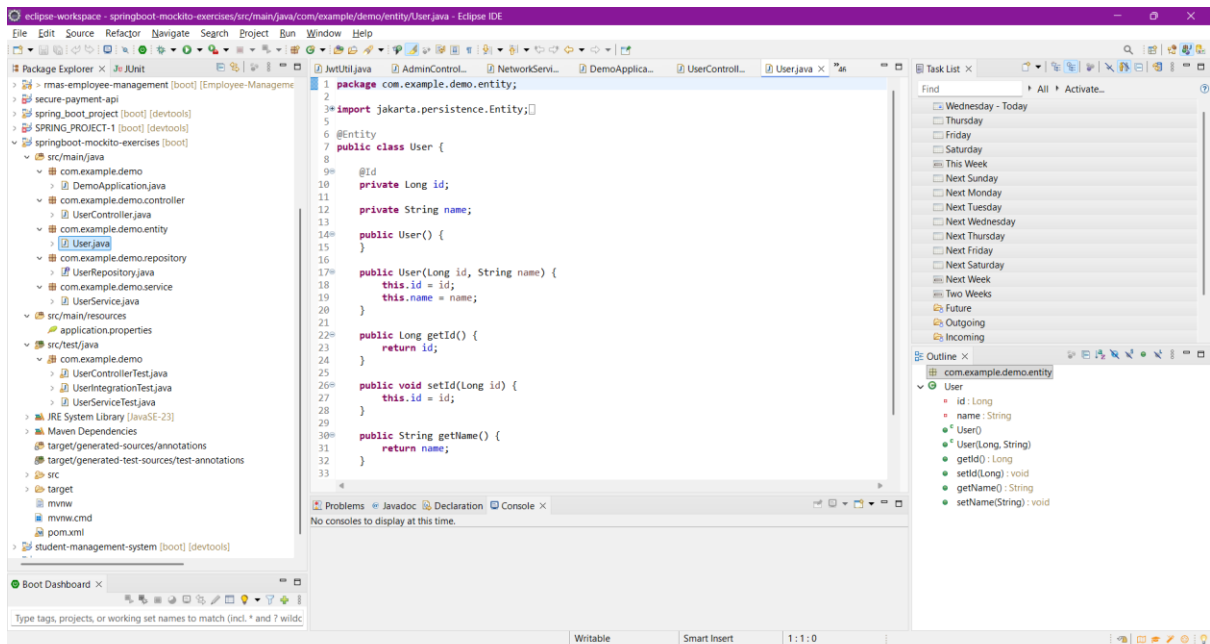
```
            .andExpect(
```

```
                jsonPath("$.name")
```

```
                    .value("John"));
```

```
    }
```

```
}
```

- SPRING_PROJECT-1 [boot] [devtools]
- springboot-mockito-exercises [boot]
 - src/main/java
 - com.example.demo
 - DemoApplication.java
 - com.example.demo.controller
 - UserController.java
 - com.example.demo.entity
 - User.java
 - com.example.demo.repository
 - UserRepository.java
 - com.example.demo.service
 - UserService.java
 - src/main/resources
 - application.properties
 - src/test/java
 - com.example.demo
 - UserControllerTest.java
 - UserIntegrationTest.java
 - UserServiceTest.java

